# Digital Ohmmeter

Grenoble INP - Ense3 / UGA
CERSE – Conception et Réalisation de Systèmes Électroniques

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## 1 Specifications Summary

The goal of this project is to design and realize a self-contained digital ohmmeter. The requirements specified for the system include:

- Measurement range: $200\,\Omega$ to $20\,\mathrm{k}\Omega$.
- Target precision: 1% nominal error margin.
- Automatic range switching using an integrated commutation mechanism.
- Local visual display using a standard $16 \times 2$ character LCD.

## 2 Objective

Measure the resistive value of an impedance and display it on an LCD screen, to be able to design a circuit following restrictions, such as a 1% error rate and find solutions using electronics components only known theoretically previously.

### 2.1 First design

The first idea for the circuit was to design a tension divisor between a potentiometer and a known resistance in series with the resistor to be measured, in which the change in the potentiometer would be the key to change the amount of current going in the other branch of resistance, for the change in current needed for higher and lower resistances, and the known resistor would be used to calculate the current, based on the known tension and resistance, so that the current could be used to calculate $R_x$.
This idea was not possible, due to the need of the potentiometer to be changed manually and the difficulty in finding a way to guarantee the precision.

## 2.2 Second Design

The second idea, was then to make a circuit maintaining the tension divisor and adding an amplifier in order to send the divided tension to the base of an also added transistor, that would make sure that the current that passes through the unknown resistor was the same one as the one that passes through a known resistor, so that it would be possible to calculate the current on the known resistance and use it to calculate the unknown.
This design did not apply well for every value of the range of resistances because the known resistance did not always have enough tension to let a reasonable amount of current pass, and that interfered when the unknown resistance was too low or too high.

## 2.3 Final Design

The circuit was then optimized with two photomos that were only activated to let current go through one resistor that let enough current through to measure the unknown resistance. The functionality is explained further below.

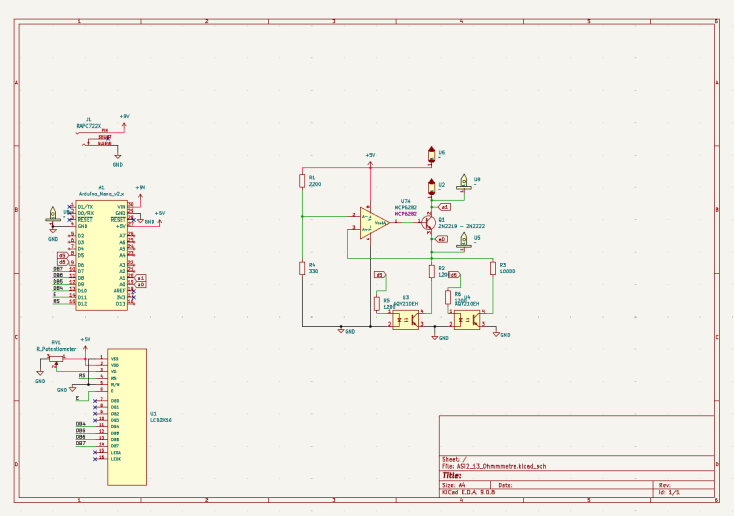


Figure 1: Scheme of the electrical circuit

# 3 Functionality

The NPN transistor holds the configuration of maintaing the current on the emissor the same as the sum of the colector and base:

$$I_e = I_c + I_b$$

As the current on the base is very low, we can assume that the current that passes through the unknown resistance has the same value as the one that passes through $R_2$ or $R_3$.
The circuit always starts with the reference resistance as $R_2$, for a lower resistances range, which is used for the first calculation of the current of the circuit ($I_1$):

$$I_1 = \frac{Va0}{R_2}$$

That is then used in the upper part of the circuit:

$$V_x = 5V - Va1$$

$$R_x = \frac{5V - Va1}{I_1}$$

The choice between $R_2$ and $R_3$ for the real measurement is then made by a commutation system that, based of the $R_x$ calculated, estimates if $R_x$ is int the range of low or high resistances. If it is a low resistance, the arduino sends a digital signal from $d5$, if it's high, sends a digital signal from $d6$.
Those digital signals are the tensions needed to activate the LED inside the photomos, that closes the circuit (as a mosfet) and lets current flow through $R_2$ (for $R_x$ = lower resistances) or $R_3$ (for $R_x$ = higher resistances).
By letting current flow, the new current is taken by the calculation of $R_x$, and a more precise measure is made.
This circuit is designed to work in a range of resistances between 200 Ohms and 20 kOhms with a 1% error rate.

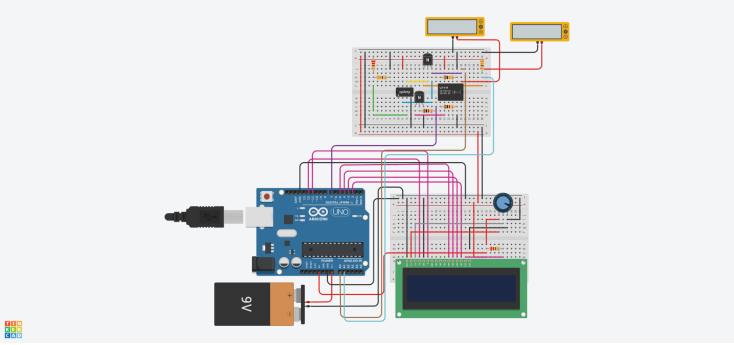



Figure 2: TinkerCad Circuit

**TinkerCad Simulation Link:** `https://shorturl.at/qMHuZ`

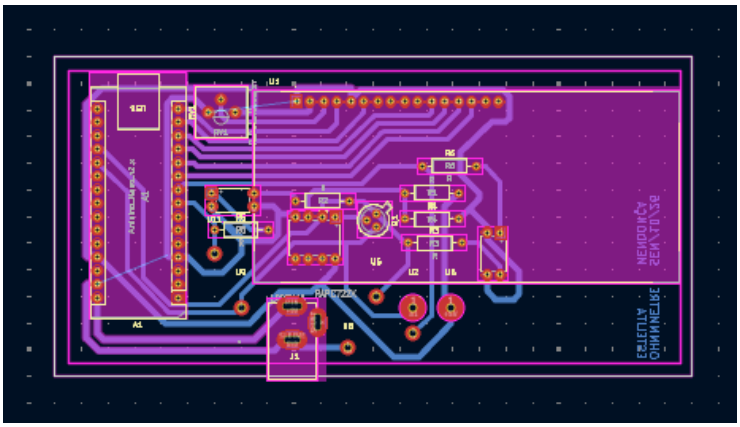


Figure 3: Kicad PCB scheme

# 4 Results

Due to external issues in the process of printing the PCB, it was not possible to guarantee the 1% error rate precision. During the tests on the breadboard, it was possible to see the whole functionality of the circuit with issues only for resistances with a value lower than 330 Ohms, which were measured with an error higher than expected.
When the circuit was then transfered to the PCB, due to inaccuracies with extra wires and reconnections on the amplifier, the measures were not in the the precision parameters.

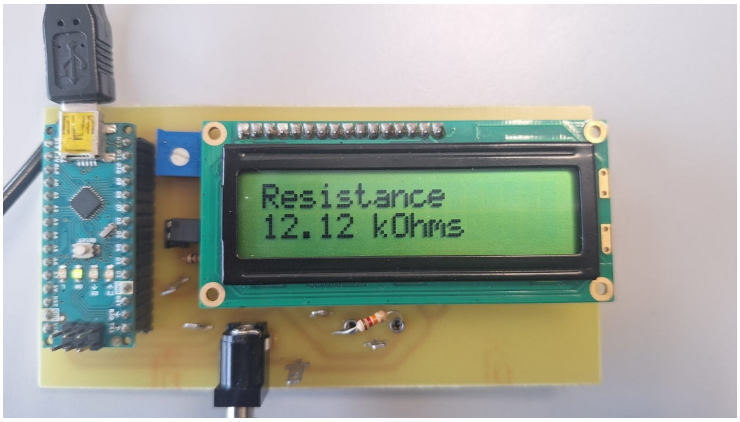



Figure 4: Functional Circuit on PCB

Table 1: Expected and displayed values

| Expected Value | Displayed Value | Error |
|---|---|---|
| 270 Ω | 261,70 Ω | 3,07% |
| 380 Ω | 386,40 Ω | 1,68% |
| 470 Ω | 462,10 Ω | 1,68% |
| 560 Ω | 548,30 Ω | 2,09% |
| 680 Ω | 668,20 Ω | 1,75% |
| 820 Ω | 803,20 Ω | 2,05% |
| 1,5 kΩ | 1,48 kΩ | 1,33% |
| 2,2 kΩ | 2,15 kΩ | 2,27% |
| 3,3 kΩ | 3,22 kΩ | 2,42% |
| 4,7 kΩ | 4,59 kΩ | 2,34% |
| 5,6 kΩ | 5,47 kΩ | 2,32% |
| 6,8 kΩ | 6,82 kΩ | 0,29% |
| 8,2 kΩ | 8,00 kΩ | 2,43% |
| 10 kΩ | 10,08 kΩ | 0,80% |
| 12 kΩ | 12,12 kΩ | 1,00% |
| 15 kΩ | 14,63 kΩ | 2,47% |
| 18 kΩ | 18,33 kΩ | 1,83% |
| 20 kΩ | 19,51 kΩ | 2,45% |

# 5 Problems and Improvements

## 5.1 Encountered Problems and Troubleshooting

- **PCB printing and jumpers:** Faulty trace printing necessitated manual wire jumpers across the board, which introduced parasitic resistance and voltage drops.
- **Low Resistance Nonlinearity:** Below 330 Ω, contact resistances and the internal ON-resistance of the photomos relay affected the tension divisor accuracy.

## 5.2 Possible Improvements

- **Production of another PCB:** Guaranteeing that every pin in well connected to the circuit.
- **Adding another Photomos:** The tension on the photomos were very close to the minimum before the saturation, the possibility of adding another photomos with a third resistance would give the project a larger margin of error to be comfortable with any measurement.

# 6 Flowchart of the code

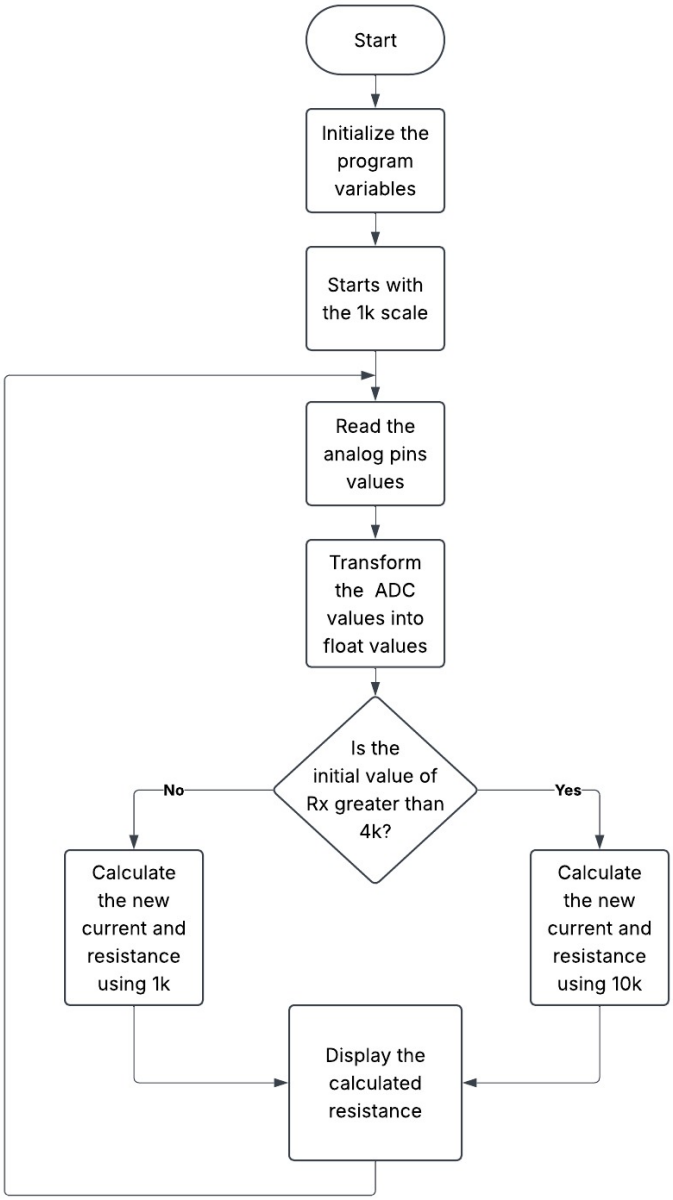



Figure 5: Flowchart

# 7 Code in C++

```
#include <LiquidCrystal.h>

const int rs = 12, en = 11, d4 = 10, d5 = 9, d6 = 8, d7 = 7;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);

const int pinoA0 = A0;
const int pinoA1 = A1;
const int pino1k = 5;
const int pino10k = 6;
bool escala10kAtiva = false;
bool escala1kAtiva = false;

void setup() {
  lcd.begin(16, 2);
  pinMode(pino1k, OUTPUT);
  pinMode(pino10k, OUTPUT);

  // Inicia na escala de 1k
  digitalWrite(pino1k, HIGH);
  digitalWrite(pino10k, LOW);
  escala10kAtiva = false;
  escala1kAtiva = true;
}

void loop() {
  if (escala1kAtiva) {
    digitalWrite(pino1k, HIGH);
    digitalWrite(pino10k, LOW);
    escala10kAtiva = false;
  } else if (escala10kAtiva) {
    digitalWrite(pino10k, HIGH);
    digitalWrite(pino1k, LOW);
    escala1kAtiva = false;
  }
  delay(35);

  analogRead(pinoA0);
  analogRead(pinoA1);
  delay(5);

  long somaA0 = 0, somaA1 = 0;
  for (int i = 0; i < 50; i++) {
    somaA0 += analogRead(pinoA0);
    somaA1 += analogRead(pinoA1);
    delay(2);
  }

```

```
  float adcA0 = somaA0 / 50.0;
  float adcA1 = somaA1 / 50.0;

  float VA0 = (adcA0 * 5.0) / 1023.0;
  float VA1 = (adcA1 * 5.0) / 1023.0;

  // Seleciona a resistencia de referencia
  float resistenciaReferencia = escala10kAtiva ? 10000.0 :
      1000.0;

  float I_circuito = VA0 / resistenciaReferencia;

  float VRx = 5.0 - VA1;
  if (VRx < 0) VRx = 0;

  float Rx = 0;
  if (I_circuito > 0.00001) {
    Rx = VRx / I_circuito;
  }

  if (escala10kAtiva == false && Rx >= 4000.0 ) {
    escala10kAtiva = true;
    escala1kAtiva = false;
    return;
  }
  else if (escala1kAtiva == false && Rx < 4000.0) {
    escala1kAtiva = true;
    escala10kAtiva = false;
    return;
  }

  lcd.setCursor(0, 0);
  lcd.print("Resistance       ");
  lcd.setCursor(0, 1);

  if (Rx >= 1000.0) {
    lcd.print(Rx / 1000.0, 2);
    lcd.print(" kOhms        ");
  }
  else {
    lcd.print(Rx, 1);
    lcd.print(" Ohms          ");
  }

  delay(400);
}
```